Signed graph union mutation invariant

Five main declarations and eight controls passed fresh Lean replay. Thirteen named axiom readbacks contain only standard Lean axioms. A deliberately false equality returned native and gate exit one.

Global preservation. The model stores the full parent/potential map and fixed anchor pairs. Every vertex equation and stored anchor equation is preserved when the native union changes the child parent and potential and conditionally transfers its fixed anchor. Both union directions derive the endpoint equation from the reduced edge and unit-square signs. Descendants and the retained child anchor are covered.

Both fixed. Existing parent anchor values remain unchanged. The difference constraint returns the actual k for nonzero target coefficient. A zero target coefficient entails a zero constant, matching the native assertion.

Controls. A three-vertex signed descendant example checks global equations, descendant frame, transferred and stale anchors, both-fixed answer two and reverse update. A wrong identity potential breaks the invariant.

Boundary. This is a post-find field-state proof. Incoming edges and prestate equations are premises. Acyclicity and size preservation under union, allocation, native word/array/counter semantics and compiled Rust correspondence remain separate. Cached dependencies are trusted. Independent review follows publication; no scientific execution occurred.

Retained failure. An initial substituted-binder failure remains in its unique invocation directory. Final acceptance checks exact module order, native exits, unchanged bindings and all thirteen readback names.